

Rahul Inturi

Inturirahul75@gmail.com

+1(425)6382488

PROFILE

Dynamic and detailed-oriented professional with a bachelor's degree in Automobile Engineering and a master's degree in computer science. Possesses a robust foundation in automotive systems, including engine dynamics and vehicle mechanics, complemented by advanced proficiency in software development, data structures, and machine learning. Demonstrated ability to integrate mechanical engineering principles with computational technologies to innovate and optimize automotive solutions. Eager to contribute to cutting-edge projects at the intersection of automotive engineering and computer science.

EXPERIENCE

INTERN, AUTONXT, Hyderabad

JAN 2023 -FEB 2023

- Acquired foundational knowledge of automobile systems including engine components, transmission, braking and suspension systems.
- Observed and assisted in vehicle diagnostics and routine maintenance tasks.

Vehicle Dynamics Engineer in R&D, Alwis Automobiles, Nellore

SEP 2021 -AUG 2022

- Cultivated robust working relationships with clients via outstanding communication skills, promoting trust and teamwork.
- Wrote, reviewed and edited technical documents in accordance with template requirements.
- Developed positive working relationships with stakeholders to effectively coordinate work activities.
- Achieved successful project outcomes by maintaining accurate documentation and meeting strict deadlines.

WORKSHOP:

Design & Fabrication Engineer – Automotive Projects

Tools: CATIA V5, 3D CAD, DFMEA, FEA, GD&T

- Successfully designed and developed multiple vehicle components, including vehicle frames and battery hold-down systems, using CATIA V5 and 3D CAD tools.
- It played a key role in the fabrication and assembly of Go-Kart vehicles, contributing to improved team performance in competitive racing events.
- Gained practical insights into vehicle dynamics and performance optimization through active involvement in hands-on construction and tuning.
- Collaborated closely with cross-functional teams to ensure material availability, minimizing delays and preventing product repeatability issues during M-BAJA vehicle development.
- Implemented effective strategies to overcome manufacturing challenges, resulting in the timely completion and successful delivery of high-quality M-BAJA vehicles.
- Designed plastic components and integrated chassis assemblies for Go-Karts, with experience in basic lathe machining tasks to support custom fabrication needs.
- Proficient in Design Failure Mode and Effects Analysis (DFMEA), Design Verification Plan and Report (DVP&R), Finite Element Analysis (FEA), and Geometric Dimensioning & Tolerancing (GD&T).

Technical skills:

Ansys, AutoCAD, SolidWorks, and Autodesk Fusion 360 for engineering design, simulation, and drafting.

Coding Skills:

Python, Power BI, Microsoft Excel, Basic Java, Basic UI/UX

Certifications

Intern

- Auto NXT
- Alwis Automobiles

Projects

Library Management System:

Technologies: Python, Django, MySQL, HTML, CSS, Bootstrap, JavaScript

- Designed and developed a full-stack Library Management System to streamline book lending and administrative workflows.
- Implemented a role-based login system with separate Admin and Student dashboards for secure and personalized access.
- Built a dynamic Event Gallery Module for admins to showcase events, workshops, and lectures with responsive UI elements.
- Developed a book reservation system with availability checks and conflict resolution for scheduled bookings.
- Automated inventory management, fine calculation, and overdue tracking, reducing manual work by 50%.
- Enabled real-time search and filter functionality by title, author, and category to improve user experience.
- Integrated full CRUD operations for books, users, and events with robust error handling and rollback mechanisms.
- Implemented a notification system using Django messages for overdue alerts and reservation confirmations.
- Built a responsive, accessible UI using Bootstrap and JavaScript, compatible across devices.
- Used Django ORM with MySQL to ensure efficient data management and optimized query performance.
- Conducted unit testing and scenario-based validation to ensure stability and handle edge cases like login failures or overlapping reservations.
- Showcased the project in academic evaluations and tech events; received positive feedback for functionality and real-world relevance.

Design and Impact Analysis of Bumper Beam and Crash Box

- The crash analysis was performed of bumper beam and crash box sub frame for automotive vehicles. The frontal crash of the integrated car system is successfully simulated and analyzed in ANSYS software.
- According to the latest bumper beam and crash box materials like polypropylene and aluminum alloy and structural steel.
- We involved two different materials like glass fiber epoxy and kenaf materials, the specimen is involved testing methods of tensile compression and impact test, the GFRP has more stress and strain characteristics compared to structural steel.
- The structural analysis will perform, and the results will compare.

EDUCATION

Master Of Science – Computer Science

APR 2023 – DEC 2024

University Of Bridgeport, Bridgeport

- Relevant coursework: Python, Deep Learning, OOPS

Bachelor of Engineering (B.E), Automobile Engineering

July 2017 – MAY 2021

Hindustan University, Chennai

- Relevant coursework: Vehicle Dynamics, Automotive Electronics, Automotive Transmission.